Malware, short for malicious software, refers to any program or code designed to disrupt systems, damage data, or gain unauthorised access to devices and networks. It is one of the most common tools used in cyber attacks and can affect both individuals and organisations. Malware can operate in the background without detection, making it a significant threat to digital security.

Malware is typically delivered through methods such as email attachments, malicious downloads, infected websites, or compromised software. In many cases, it is combined with other attack techniques, such as phishing or social engineering, to increase the likelihood of success. Once installed, malware can perform a range of harmful actions depending on its purpose.

There are several common types of malware, each with different characteristics:

- **Viruses** – attach themselves to files or programs and spread when those files are executed
- **Worms** – self-replicating malware that spreads across networks without user interaction
- **Trojans** – disguised as legitimate software but contain malicious functionality
- **Ransomware** – encrypts files or systems and demands payment for their release
- **Spyware** – secretly collects information about user activity
- **Adware** – displays unwanted advertisements and may track user behaviour

These types are explored in more detail in other sections of the site, where their behaviour and impact are examined further.

Malware infections typically follow a process. A user may unknowingly download or open a malicious file, or visit a compromised website. Once executed, the malware installs itself on the device and may attempt to remain hidden. It can then begin carrying out its intended function, such as stealing data, monitoring activity, spreading to other systems, or disrupting normal operations.

For individuals, malware can lead to data loss, unauthorised access to personal accounts, financial fraud, and reduced device performance. In some cases, sensitive information such as login credentials or banking details may be captured without the user's knowledge.

For businesses, the impact is often more severe. Malware infections can result in large-scale data breaches, operational downtime, financial losses, and damage to reputation. Ransomware attacks, in particular, can halt business operations entirely until systems are restored or demands are addressed.

Recognising malware can be difficult, but there are several common indicators:

- Devices running slower than usual or behaving unexpectedly
- Frequent crashes or error messages
- Unfamiliar programs or files appearing on the system
- Excessive pop-ups or unwanted advertisements
- Unusual network activity or data usage
- Security software being disabled without user action

While these signs do not always confirm malware, they may indicate a potential issue that requires investigation.

Protecting against malware involves a combination of user awareness and technical controls:

- Avoiding downloads from untrusted or unknown sources
- Being cautious with email attachments and links
- Keeping operating systems and software up to date
- Using reputable security software such as antivirus or endpoint protection
- Regularly backing up important data
- Limiting user permissions to reduce the impact of potential infections

For organisations, additional measures such as network monitoring, access controls, and employee training can further reduce the risk of infection and spread.

Malware remains a persistent threat because it continues to evolve in both complexity and delivery methods. Attackers regularly adapt their techniques to bypass security controls and exploit new vulnerabilities.

This page provides an overview of malware, including its types, how it spreads, and its potential impact. Understanding how malware operates and recognising the signs of infection are important steps in maintaining secure systems and protecting sensitive information.